

orbit_demo physical design (sky130hd, variant sep)

Package section 06_physical_design of orbit_demo_design_review, written 2026-10-06. Links are relative to this file. Paths in backticks that start with build/ or viz/ are repository paths (repo /home/user/chip); they are not in the package unless the text links a package copy. The pd/, scripts/ and mk/ files cited here are copied byte-identical under 04_digital_design/source/, except scripts/viz_parts.py (repository only); reports/pdsep/ is copied under 07_verification/published_reports/ except its GDS archive (see Related evidence below). Repository HEAD at writing: 54745f7; git diff 0495cfa 54745f7 -- rtl pd mk scripts reports docs viz is empty, so the design sources are those of the package commit in the revision table. Discrepancies are listed in the discrepancy register (09_review_notes).

Status labels: **VERIFIED** = a tool log or report shows it. **REPRODUCED 2026-10-06** = re-run for this package. **TARGET** = design goal, not measured. **ASSUMPTION. CLAIM (unverified)** = stated in a document, no log found. **MISSING. N/A** = not applicable (reason given).

This section covers only the reviewed layout (variant sep, 7.2 ns). The reference layouts are not the reviewed layout. Baseline (variant base, 7.0 ns) numbers appear only as comparison values (section 4.2 here; 07_verification/baseline_reference/). No IHP SG13G2 result is used in this section. The ORBIT-AI brief's 16-tile architecture, 200-800 MHz clock target and 40 W sizing assumption describe a different, unbuilt chip and have no counterpart in this layout.

Design revision covered by this package

Item	Revision / identifier	Note
Design	orbit_demo (ORBIT-AI 4-lane INT8 digital demonstrator)	top module orbit_demo, LANES=4
RTL	rtl/orbit_demo.v, orbit_mac_lane.v, orbit_thermal_tmr.v, orbit_keep_reg.v at git commit 26566f0 (2026-09-29 18:54 UTC); unchanged through the package commit	sha256 a3fffb22... / efef3d33... / bba77a5a... / ea05bdc1... (full hashes in MANIFEST.sha256 at the package root)
Specification	docs/SPEC.md at 26566f0 (never revised)	sha256 aef5d9fb...
Concept brief	docs/orbit-ai-design-brief.pdf, "ORBIT-AI v0.1, 29 September 2026" (never revised; describes an earlier project state, see discrepancy register)	sha256 9d5f33d3...
Reviewed layout	ORFS sky130hd, variant sep (copy-separation fences), run 2026-09-29 21:56-22:12 UTC, outputs build/pd_sep/results/sky130hd/orbit_demo/sep/	6_final.gds sha256 cd19afa9...; 6_final.def f5c544f3...; 6_final.v 68093c41...; 6_final.spef 014e655a...
Layout configuration	pd/sky130hd_sep/config.mk and regions.tcl at c10c59b; constraint.sdc at 21917cf (set clk_period 7.2)	The run (21:56 UTC) used config.mk/regions.tcl as in ed553f8, identical to c10c59b once comments are stripped. It used the working-tree constraint.sdc with period 7.2 ns, which was committed in 21917cf at 22:00 UTC and is byte-identical to the flow copy build/pd_sep/sdc/sky130hd/sep.sdc. ed553f8's constraint.sdc has a 7.0 ns period, so a rebuild must not use it
Flow / tools	ORFS docker image tag openroad/orfs:latest; the only local image is sha256:2e5bf6fe865e... (created 2026-09-29 02:00 UTC, before the run). Linking the run to this digest is an ASSUMPTION: the run logged no digest. OpenROAD prints version "unknown". KLayout 0.30.12 (DRC/LVS). Yosys 0.69+154 (local sims/schematics)	No PDK commit is recorded in the logs
Process / library	SkyWater SKY130, sky130_fd_sc_hd, liberty sky130_fd_sc_hd_tt_025C_1v80 (the single corner ORFS optimised at)	—

Item	Revision / identifier	Note
Package assembled	2026-10-06; design inputs copied from the repository tree at commit 0495cfa (branch claude/hopeful-rubin-0yf8io)	Every later commit on the branch changes only review/ or is empty (git diff --name-only 0495cfa..HEAD outside review/ is empty). Re-runs made for this package are dated 2026-10-06

Other implementations in the repo (sky130hd baseline at 7.0 ns, variant base; IHP SG13G2) are reference runs only and are NOT the reviewed layout.

Notes for this section: the image tag is set in mk/pdsep.mk line 25 and logged in build/pd_sep/make_pd_sep.out line 1 (packaged as [make_pd_sep.out.gz](#)). The ORFS commit and the open_pdk / skywater-pdk version are MISSING (not logged; the image's /0openROAD-flow-scripts is not a git repository).

Contents of this folder

Path	Content	Origin
layout_db/	final GDS, DEF, SPEF, gate netlist, OpenROAD database 6_final.odf (gzip), final SDC, clock period	copied from build/pd_sep/results/sky130hd/orbit_demo/sep/
klayout_views/	10 KLayout renders of the final GDS (v01-v08) and render_gds.py	rendered for this package, KLayout 0.30.12 Python module
orfs_images/	10 OpenROAD GUI images written by the ORFS final report step	copied from build/pd_sep/reports/sky130hd/orbit_demo/sep/
floorplan_plot/	floorplan_fences_pins.png and its script plot_floorplan.py	plotted for this package from package files only
physical_design.pdf	this document rendered to PDF	md2pdf.py README.md physical_design.pdf

Related evidence elsewhere in the package: ORFS step logs [07_verification/orfs_logs/](#) (byte-identical copies of 39 of the 40 files in build/pd_sep/logs/sky130hd/orbit_demo/sep/; the 40th, 5_1_grt.json, is in [07_verification/additional_evidence/flow/](#)), ORFS reports [07_verification/orfs_reports/](#) (all non-image report files of build/pd_sep/reports/sky130hd/orbit_demo/sep/ except drt_antennas.log and grt_antennas.log, which are 0 bytes and are in [07_verification/additional_evidence/flow/](#)), the published copy-separation reports [07_verification/published_reports/](#) (= reports/pdsep/ except orbit_demo_sky130hd_sep.gds.gz, the same GDS as [layout_db/6_final.gds.gz](#)), the pin table [08_pinout_packaging/pinout.csv](#), the ORFS design configuration [04_digital_design/source/pd/sky130hd_sep/](#), and the LVS reference netlist [03_schematics/netlists/6_final.cdl.gz](#) (with the library CDL appended: [6_final_concat.cdl.gz](#)).

1. Layout database inventory

Run: make pd-sep (ORFS, sky130hd, variant sep), files written 2026-09-29 UTC. The sha256 values are of the uncompressed originals in build/pd_sep/results/sky130hd/orbit_demo/sep/. Check made for this package: gunzip -c <file>.gz | sha256sum compared with sha256sum of the build file, and gzip -t on all five archives (6_final.odf.gz added and checked at the final step): all five match, archives intact (**REPRODUCED 2026-10-06**).

Package file	Original in build/pd_sep/results/sky130hd/orbit_demo/sep/ (mtime UTC)	Bytes .gz / uncompressed	gunzip = build	Status
6_final.gds.gz	6_final.gds (22:09:49)	1088094 / 8174720	match	REPRODUCED 2026-10-06
6_final.def.gz	6_final.def (22:09:33)	852148 / 6992153	match	REPRODUCED 2026-10-06
6_final.spef.gz	6_final.spef (22:09:39)	1956262 / 6827076	match	REPRODUCED 2026-10-06

Package file	Original in build/pd_sep/results/sky130hd/orbit_demo/sep/ (mtime UTC)	Bytes .gz / uncompressed	gunzip = build	Status
6_final.v.gz	6_final.v (22:09:33)	96856 / 812562	match	REPRODUCED 2026-10-06
6_final.odb.gz	6_final.odb (22:09:33)	2751275 / 14668665	match	REPRODUCED 2026-10-06
6_final.sdc	6_final.sdc (22:09:33)	- / 19270	byte- identical (cmp)	REPRODUCED 2026-10-06
clock_period.txt	clock_period.txt (21:56:09), content 7.2	- / 4	byte- identical (cmp)	REPRODUCED 2026-10-06

File (uncompressed original)	sha256
6_final.gds	cd19afa93221c8468567278bcd627657bd6841f5ace7e0bbe749e0c76b93bc64
6_final.def	f5c544f37701038b0cf24fdc09481c58b249c425539b229d000ccb8a9df20ec1
6_final.spef	014e655ae938245ace4ebc279b9c6c311ddf57f23bfae44367af342aeb8d5acb
6_final.v	68093c41fca50e9e532205b1128490a1e7807e63d998ffa179ca985981c8b06
6_final.odb	3a6144cb4f882f8e9f7a2acc9d52fa0375f3c313797b1c0e29050567dabac7e8
6_final.sdc	1dbe8e7e8cd2853f6b5d533496b080b2aca8bfa82e5f10ed1714f37922409680
clock_period.txt	69cd84998732b964b8b9c218f595f2aa1c438cd71d725a2542227075cd721969

What the files contain:

File	Content (checked for this package)	Source / conditions	Status
6_final.gds	top cell orbit_demo, 154 cells, dbu 0.001 um, bbox (0,0;346.295,346.295) um; identical to 6_1_merged.gds (same size)	read with KLayout 0.30.12 Python module (klayout.db)	REPRODUCED 2026-10-06
6_final.gds merge	"All LEF cells have matching GDS/OAS cells", "No orphan cells in the final layout"; standard-cell GDS sky130_fd_sc_hd.gds of the ORFS sky130hd platform	6_1_merge.log lines 1-5, KLayout 0.30.12	VERIFIED
6_final.def	DEF 5.8, UNITS DISTANCE MICRONS 1000, DIEAREA (0 0) (346295 346295), 125 ROWs, REGIONS 5, COMPONENTS 18175, PINS 217, SPECIALNETS 2, NETS 6886, GROUPS 5	decompressed lines 1-6, 7-131, 162-168, 169, 18346, 19239-27606, 27607, 126373-126769	REPRODUCED 2026-10-06
6_final.spef	IEEE 1481-1999, written by OpenROAD (*VERSION "unknown") 22:09:39; OpenRCX with platforms/sky130hd/rcx_patterns.rules, one extraction corner X, coupling threshold 0.1 fF; 6886 nets, 30095 rsegs, 60471 coupling caps	SPEF header; 6_report.log lines 16-26. Provenance/qualification of rcx_patterns.rules: MISSING	VERIFIED
6_final.v	routed gate netlist, module orbit_demo, 18 ports / 215 bits, 6250 cell instances; fill, tap and antenna-diode cells are not written; no VDD/VSS ports	zcat + count of sky130_* instance lines	REPRODUCED 2026-10-06
6_final.sdc	create_clock -name clk -period 7.2000, set_propagated_clock, 79 set_input_delay and 135 set_output_delay of 1.44 ns (0.2 x period, an ASSUMPTION of the SDC for the unknown outside world)	file lines 8-10; constraint source constraint.sdc	VERIFIED

6_final.odb (14668665 bytes, needed to re-run OpenROAD STA/PSM directly) was added to this folder on 2026-10-06 as [6_final.odb.gz](#). The LVS database 6_lvs.lvssdb and 6_final_concat.cdl (build originals in build/pd_sep/results/.../sep/ and build/pd_sep/objects/.../sep/) are in [07_verification/additional_evidence/flow/](#). The CDL 6_final.cdl is in [03_schematics/netlists/](#); the KLayout LVS extracted netlist orbit_demo_extracted.cir is in [07_verification/lvs/](#).

2. Run settings that shape the layout

Setting	Value	Source	Status
Design config	SYNTH_KEEP_MODULES = *orbit_keep_reg*; CORE_UTILIZATION 40, CORE_ASPECT_RATIO 1, CORE_MARGIN 2, PLACE_DENSITY_LB_ADDON 0.20; PLACE_PINS_ARGS = -min_distance 2 - min_distance_in_tracks; TNS_END_PERCENT 100; POST_FLOORPLAN_TCL = regions.tcl, PDSEP_FLOORPLAN edges, PDSEP_REGION_TYPE EXCLUSIVE, ENABLE_DPO 0, SLEW_MARGIN 20, CAP_MARGIN 20	config.mk lines 30, 35-42, 66-71	VERIFIED (used values appear in the logs, e.g. 2_1_floorplan.log line 16, 3_2_place_iop.log line 10)
Liberty	sky130_fd_sc_hd__tt_025C_1v80.lib only (TT, 25 C, 1.80 V)	every step log, line 6 (e.g. 2_1_floorplan.log)	VERIFIED
Tools	OpenROAD version printed as unknown (-GPU +GUI - Python); KLayout 0.30.12 (merge, DRC, LVS); Yosys 0.68+post (synthesis, git sha1 UNKNOWN)	log line 1 of each step; 1_2_yosys.log line 1	VERIFIED; OpenROAD and ORFS commits MISSING
Platform scripts	ORFS platforms/sky130hd/{pdn.tcl, tapcell.tcl, config.mk} are not logged; values quoted below from the image on this host match the DEF	read from the image layer on this host 2026-10-06	ASSUMPTION (same image as the run)

3. Floorplan

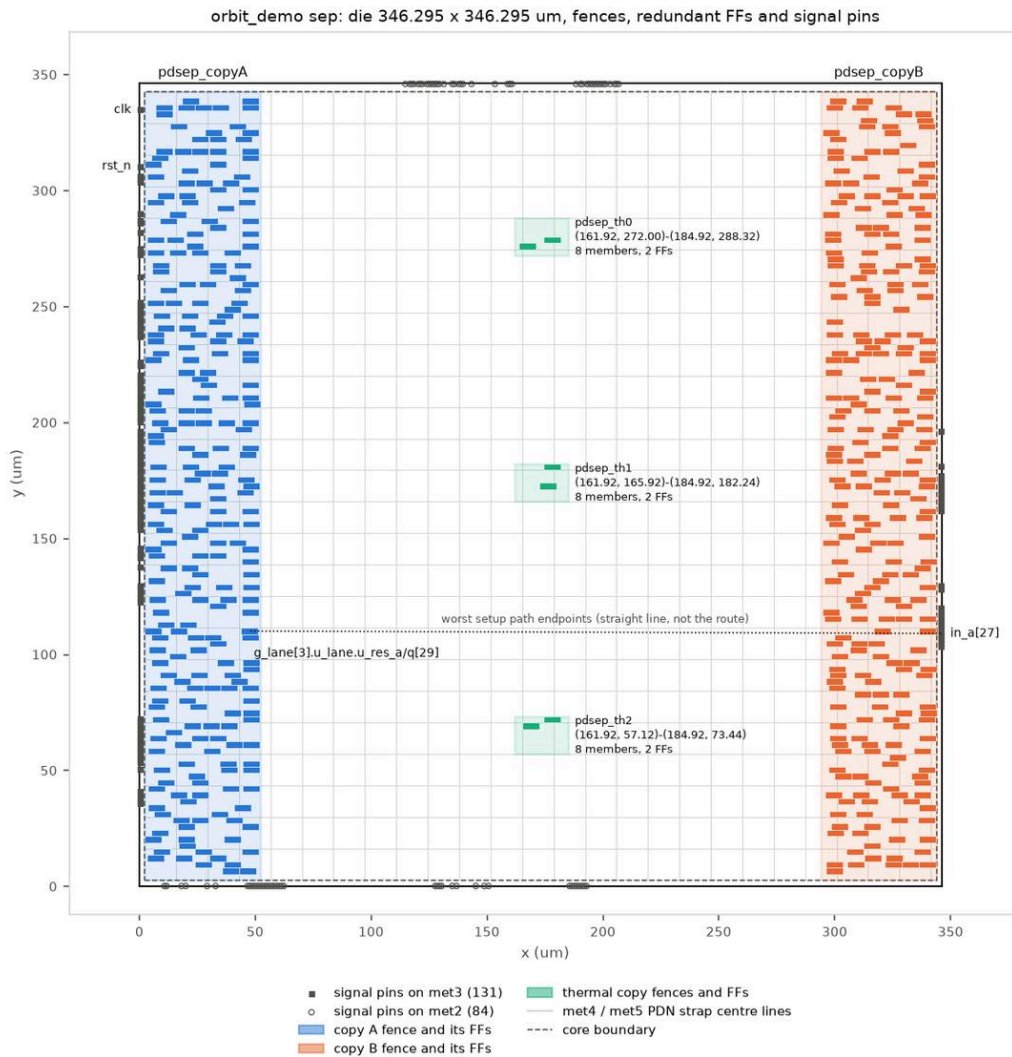


Figure F1. Source: [plot_floorplan.py](#) (matplotlib, rvenv Python 3.11) reading [regions.json](#) (die, core, fence boxes, the 518 redundant flip-flop cell boxes from the routed DEF), [pinout.csv](#) (215 signal pin origins and layers) and [6_final.def.gz](#) SPECIALNETS (met4/met5 strap centre lines). The dotted line only joins the two endpoints of the worst setup path; it is not the route.

Floorplan command: initialize_floorplan -utilization 40 -aspect_ratio 1 -core_space 2 -site unithd (2_1_floorplan.log line 16).

Item	Value	Source	Status
Die	(0, 0) - (346.295, 346.295) um, 119920 um ²	2_1_floorplan.log line 20; DEF line 6	VERIFIED
Core	(2.300, 2.720) - (344.080, 342.720) um = 341.780 x 340.000 um, 116205.200 um ² ; lower-left snapped from (2.000, 2.000) (IFP-0028)	2_1_floorplan.log lines 18, 21-22	VERIFIED
Rows	125 rows x 743 sites unithd (site 0.46 x 2.72 um), alternating N / FS, x 2.30-344.08 um	2_1_floorplan.log line 19 (IFP-0001); DEF ROW lines 7-131 (STEP 460, row pitch 2720)	VERIFIED
Macros	none ("Found 0 macro blocks")	3_2_place_iop.log line 11	N/A: standard-cell-only block
Utilization	target 40 % (config); effective 0.403 at floorplan (46866.198 um ² synthesis cell area); 41 % after floorplan repair_timing; 43 % after tap cells; final 54 % (Design area 62300 um ² , i.e. cell area excluding fill, including tap: 62299.8 um ² / 116205.2 um ² = 0.536)	2_1_floorplan.log lines 24, 398; 2_3_floorplan_tapcell.log line 11; 6_report.log line 67; results.md lines 42-44	VERIFIED
Placement density	global_placement -skip_io: 0.53 = 0.40 lower bound + (1 - 0.40) x PLACE_DENSITY_LB_ADDON 0.20 + 0.01; main (timing- and routability-driven) global placement: 0.546 (lower bound 0.42)	3_1_place_gp_skip_io.log lines 11-12; 3_3_place_gp.log lines 15-16; formula from ORFS flow/scripts/util.tcl line 198 in the image on this host	VERIFIED (values); formula ASSUMPTION (same script as the run)
Tap cells	1524 sky130_fd_sc_hd__tapvpwrvrnd_1, inserted before placement	2_3_floorplan_tapcell.log line 10 (TAP-0005); DEF COMPONENTS	VERIFIED
Tap arrangement	123 rows with 12 taps at 27.6 um pitch, alternate rows offset by 13.8 um (checkerboard); first and last row 24 taps at 13.8 um; platform tapcell -distance 14; conditions: arrangement measured on the DEF; script value from the current image	DEF COMPONENTS (counted per row); platforms/sky130hd/tapcell.tcl line 2 in the image	REPRODUCED 2026-10-06 (arrangement); ASSUMPTION (script)
Filler cells	10381 (53905.45 um ²): fill_8 3465, fill_1 2521, fill_2 2369, fill_4 2026; conditions: filler_placement with fill_1/2/4/8 only	5_3_fillcell.log lines 10-11; 6_report.log line 52; DEF COMPONENTS	VERIFIED
Decap cells	0. The filler list contains no decap master; sky130_ef_sc_hd__decap_12 "is loaded but not used in the design". results.md labels the 10381 filler cells "fill/decap"; see discrepancy register (09_review_notes). Decoupling capacitance budget: MISSING	5_3_fillcell.log line 10; 2_1_floorplan.log line 10; DEF COMPONENTS (no decap master)	VERIFIED
Metal fill / density	not run: step 6_1_fill only copies the ODB (exec cp ... 5_route.odb ... 6_1_fill.odb); no density rules in the KLayout deck. The tech LEF sky130_fd_sc_hd.tlef lists MAXIMUMDENSITY 70 (window 700 x 700, step 70) for met1-met4 (image copy, ASSUMPTION same file as the run); no step checks it	6_1_fill.log line 10	VERIFIED (not run); fill and density check MISSING
Other physical cells	20 diode_2 antenna-repair cells; 6 conb_1 tie cells (RSZ-0042)	6_report.log lines 51-61; 2_1_floorplan.log line 31	VERIFIED

Final cell report (6_report.log lines 51-61, **VERIFIED**):

Cell type	Count	Area (um^2)
Fill cell	10381	53905.45
Tap cell	1524	1906.83
Antenna cell	20	50.05
Clock buffer	88	1672.85
Timing Repair Buffer	1201	9959.55
Inverter	171	648.12
Clock inverter	8	127.62
Sequential cell	521	10430.00
Multi-Input combinational cell	4261	37504.72
Total	18175	116205.20

3.1 Power distribution network

One voltage domain CORE (VDD/VSS). [2_4_floorplan_pdn.log](#) records only [INFO PDN-0001] Inserting grid: grid (line 12) and two PDN-1051 warnings for the empty macro grids (lines 10-11), so the geometry below is measured from the DEF SPECIALNETS ([6_final.def.gz](#), decompressed lines 19239-27606). The values agree with platforms/sky130hd/pdn.tcl lines 21-26 in the image on this host (ASSUMPTION that the run used the same script).

Layer	Use	Width (um)	Pitch per net (um)	Position / offset	VDD / VSS shapes	Source	Status
met1	followpin rails on every row boundary	0.48	5.44	x 2.30-344.08 um	63 / 63	DEF SPECIALNETS (SHAPE FOLLOWPIN)	REPRODUCED 2026-10-06
met4	vertical straps	1.60	27.14	VSS from x 15.87 um (core x0 + 13.57), VDD from x 29.44 um; VDD-VSS 13.57 um	12 / 13	DEF SPECIALNETS (SHAPE STRIPE)	REPRODUCED 2026-10-06
met5	horizontal straps; also the VDD/VSS pins	1.60	27.20	VSS from y 16.32 um (core y0 + 13.6), VDD from y 29.92 um; x 2.30-344.08 um	12 / 13	DEF SPECIALNETS and PINS (VDD/VSS + LAYER met5, decompressed lines 18347-18376)	REPRODUCED 2026-10-06
via stack met1-met4	via/via2/via3 arrays 1.6 x 0.48 um at every followpin x met4 crossing	-	-	-	756 / 819 of each via level	DEF SPECIALNETS (via2_3_1600_480_1_5_320_320, via3_4_1600_480_1_4_400_400, via4_5_1600_480_1_4_400_400)	REPRODUCED 2026-10-06

Layer	Use	Width (um)	Pitch per net (um)	Position / offset	VDD / VSS shapes	Source	Status
via4 met4- met5	1.6 x 1.6 um at every met4 x met5 crossing of the same net	-	-	-	144 / 169	DEF SPECIALNETS (via5_6_1600_1600_1_1_1600_1600)	REPRODUCED 2026-10-06

Static IR drop of this grid (OpenROAD PSM analyze_power_grid, corner default, VDD 1.80 V / VSS 0.0 V, total power 5.21e-02 W from report_power with the tool's default switching activity, TT 25 C, OpenRCX parasitics, 7.2 ns): VDD worst 4.54e-04 V (0.454 mV), average 9.58e-05 V; VSS worst 6.28e-04 V (0.628 mV), average 8.72e-05 V; 0.03 % each; "All shapes on net VDD/VSS are connected". Source [6_report.log](#) lines 27-50 (**VERIFIED**); same values from a PSM re-run on 6_final.odb ([ir.log](#), [ir.tcl](#); **REPRODUCED 2026-10-06**). The activity is not a workload activity (ASSUMPTION of the tool default). The re-run also computed PSM EM currents at the same default activity (-enable_em): maximum resistor current VDD 4.05e-04 A, VSS 4.33e-04 A ([ir.log](#) lines 17-23, 36-42; REPRODUCED 2026-10-06). The tech LEF read by the run, sky130_fd_sc_hd.tlef ([1_synth.log](#) line 7), contains DCCURRENTDENSITY/ACCURRENTDENSITY limits for mcon, met1-met5 and via-via4 (e.g. met1 DCCURRENTDENSITY AVERAGE 2.8 mA/um; read from the image copy on this host, ASSUMPTION same file as the run). The comparison of the PSM currents with these limits: MISSING. Dynamic IR drop: MISSING (needs switching activity from simulation).

3.2 IO pin placement

The layout is a core block: there is no pad ring, IO cells, ESD structures, bond pads or package (N/A: not designed; see 08_pinout_packaging). Pin locations are tool output, not a specified pinout: global_placement -skip_io ran first, then place_pins, and no pin constraint exists in pd/, scripts/ or mk/. A pinout specification is MISSING.

Item	Value	Source	Status
Command	place_pins -hor_layers met3 -ver_layers met2 -min_distance 2 -min_distance_in_tracks (layers from the platform defaults IO_PLACER_H met3, IO_PLACER_V met2)	3_2_place_iop.log line 10; config.mk line 40	VERIFIED
Pins	215 signal I/O, all with sinks, 1254 available slots, I/O nets HPWL 34567.11 um	3_2_place_iop.log lines 12-18	VERIFIED
DEF pins	217 = 215 signal + VDD (USE POWER) + VSS (USE GROUND); VDD/VSS pins are the met5 straps (FIXED)	DEF line 18346 PINS 217 ; and PINS section	VERIFIED
Edges and layers	W 102 and E 29 on met3 (shape 0.8 x 0.3 um); S 42 and N 42 on met2 (shape 0.14 x 0.485 um); clk at W (0.400, 334.900) um	pinout.csv ; DEF lines 18382-18393	REPRODUCED 2026-10-06
Minimum pin pitch	1.36 um on W/E (2 met3 tracks of 0.68 um), 0.92 um on N/S (2 met2 tracks of 0.46 um)	computed from pinout.csv ; DEF TRACKS lines 136-139	REPRODUCED 2026-10-06

The worst setup path starts at in_a[27] on the E edge (345.895, 109.140) um and ends at g_lane[3].u_lane.u_res_a/q[29] inside the copy-A fence (cell box x 43.70-51.06 um, y 108.80-111.52 um in regions.json; Figure F1; [worst_setup_path.txt](#)). Full pin table and lane/bit mapping: [08_pinout_packaging/pinout.csv](#).

4. Copy-separation fences

Mechanism ([regions.tcl](#), run as POST_FLOORPLAN_TCL at the end of the floorplan step): one OpenDB dbRegion (one box, snapped to the site grid and row boundaries) and one dbGroup per copy, region type EXCLUSIVE, written to the DEF as TYPE FENCE. Members are selected by instance name: ^g_lane\[0-9]+\]\.u_lane\.u_(acc|res)_a/ (copy A), ..._b/ (copy B), ^u_thermal\.u_copyK/ (thermal copies, lines 77-83); the script stops the flow if a group's flip-flop count differs from 256/256/2/2/2 (lines 85, 170-175). improve_placement is disabled (ENABLE_DPO 0) because it does not keep the fences legal (published

[mechanism_test.txt](#); test run on the baseline's 3_2_place_iop.odb, not on this run). Everything else (multipliers, adders, comparators, voter, control, clock tree, port buffers) is unconstrained.

4.1 Regions

From [regions.json](#) (generator [scripts/pdsep_separation.py](#), source [build/pd_sep/results/sky130hd/orbit_demo/sep/6_final.def](#)); boxes agree with DEF REGIONS (decompressed lines 162-168) and the floorplan log ([2_1_floorplan.log](#) lines 400-404). **VERIFIED**; regions.json re-generated byte-identical on 2026-10-06 (**REPRODUCED 2026-10-06**).

Region (group)	Holds	Box (x0, y0) - (x1, y1) um	Size um	Members at floorplan	Members in routed DEF	Flip-flops
pdsep_copyA (pdsep_grp_copyA)	copy A of every accumulator/result pair: u_acc_a + u_res_a, lanes 0-3	(2.30, 2.72) - (52.44, 342.72)	50.14 x 340.00	824	768	256
pdsep_copyB (pdsep_grp_copyB)	copy B: u_acc_b + u_res_b, lanes 0-3	(293.94, 2.72) - (344.08, 342.72)	50.14 x 340.00	824	768	256
pdsep_th0 (pdsep_grp_th0)	thermal state copy u_thermal.u_copy0	(161.92, 272.00) - (184.92, 288.32)	23.00 x 16.32	8	8	2
pdsep_th1 (pdsep_grp_th1)	u_thermal.u_copy1	(161.92, 165.92) - (184.92, 182.24)	23.00 x 16.32	8	8	2
pdsep_th2 (pdsep_grp_th2)	u_thermal.u_copy2	(161.92, 57.12) - (184.92, 73.44)	23.00 x 16.32	8	8	2

The 824 -> 768 change per copy group is the synthesis buffers removed before global placement ("Removed 182 buffers", [3_3_place_gp.log](#) line 10); the routed DEF GROUPS section holds 768/768/8/8/8 members (counted for this package). Copy-A/B members: the 256 kept flip-flops plus 256 mux2i_1 and 256 nor2b_1 enable/reset gates. Thermal members: 2 dfxtp_1, 2 mux2i_1, 1 nor2b_1, 1 nand2_1 and the 2 conb_1 tie cells inserted by repair_tie_fanout (RSZ-0042, [2_1_floorplan.log](#) line 31, before regions.tcl runs at line 399). Resizer and CTS buffers are not members.

Fence edge-to-edge gaps ([separation.md](#), region-pair table): copyA/copyB 241.50 um; copyA to th0/th1/th2 109.48 um; copyB to th0/th1/th2 109.02 um; th0/th1 89.76 um; th1/th2 92.48 um; th0/th2 198.56 um (**VERIFIED, REPRODUCED 2026-10-06**).

4.2 Separation metrics

Measured by [scripts/pdsep_separation.py](#) on the final routed DEFs: separated = this layout, baseline = [build/pd/results/sky130hd/orbit_demo/base/6_final.def](#) (the unconstrained 7.0 ns reference run, not in the package). Cell boxes = DEF origin + LEF SIZE from [build/pd_sep/platform/cells.lef](#); centre = centre-to-centre, gap = edge-to-edge. Geometry only: no radiation transport, charge-collection or upset-rate model. Published results: [separation.md](#), [separation.json](#), [summary.md](#) lines 5-10. Re-run 2026-10-06 (Python 3.11.15, command from [mk/pdsep.mk](#) lines 90-94): separation.md, separation.json and regions.json byte-identical to the published files, "acceptance check (separated): PASS".

Quantity	Baseline	Separated	Acceptance target	Status
acc/res same-bit centre distance, min	2.76 um	251.17 um (lane1.res bit 25, lane3.res bit 10)	>= 20 um (TARGET)	REPRODUCED 2026-10-06
acc/res same-bit centre distance, median of group medians	30.70 um	292.57 um	-	REPRODUCED 2026-10-06

Quantity	Baseline	Separated	Acceptance target	Status
min edge gap, any copy-A FF vs any copy-B FF	0.00 um	242.94 um	>= 10 um (TARGET)	REPRODUCED 2026-10-06
copy fence edge gap (pdsep_copyA / pdsep_copyB)	no fences	241.50 um	-	REPRODUCED 2026-10-06
thermal same-bit centre distance, min / median	5.52 / 8.44 um	97.92 / 106.26 um	>= 20 um (TARGET)	REPRODUCED 2026-10-06
thermal min edge gap between any two copies (FFs)	0.00 um	92.54 um	>= 10 um (TARGET)	REPRODUCED 2026-10-06
thermal fence gap, min	no fences	89.76 um (th0/th1)	>= 10 um (TARGET)	REPRODUCED 2026-10-06
same-bit pairs in touching or overlapping cells	13 of 262	0 of 262	0 (TARGET)	REPRODUCED 2026-10-06
group members outside their region	n/a	0	0 (TARGET)	REPRODUCED 2026-10-06
non-member cells fully inside a fence / straddling a fence edge (excl. fill, tap)	n/a	2 (diode_2, antenna repair after placement) / 107	none set	REPRODUCED 2026-10-06

Notes:

- The acceptance targets (0 touching, >= 20 um centre, >= 10 um gap, every member inside) are hard-coded in [scripts/pdsep_separation.py](#) lines 24-33 and 46-48 with no stated rationale; the brief gives no distance. They are TARGETS; a physical basis (sky130 multi-node-upset / charge-sharing data) is MISSING.
- The "242 um" copy distance quoted in [summary.md](#) lines 151 and 153 is the fence gap 241.50 um printed with 0 decimals: [scripts/pdsep_summary.py](#) computes it from the region boxes (line 137, copy-B x0 minus copy-A x1) and prints it at line 445. It is not the 242.94 um FF-to-FF gap, which would print as 243. regions.tcl says "about 240 um"; see discrepancy register (09_review_notes).
- The second, older measurement script [pd/copy_separation.py](#) agrees ([copy_separation_pd_script.txt](#): res min 251.17 um, acc min 252.55 um, thermal min 97.92 um), but its line 2 "Placement was NOT constrained" is fixed text written for the baseline; see discrepancy register (09_review_notes).
- Common-mode logic is not separated: one clock tree (TritonCTS, net clk, 521 sinks, one tree, [4_1_cts.log](#) lines 17-18, 43-54), the rst_n tree, the shared 8x8 multiplier of each lane, the A/B comparators and mismatch OR tree, the thermal voter and next-state logic, the port buffers and the single flip-flops u_thermal.phase, out_valid_q, fault_q (list in regions.json unconstrained_common_mode; [summary.md](#) lines 192-200). The clock root buffer clkbuf_0_clk overlaps fence pdsep_th1 by 95 % of its area ([summary.md](#) line 123). Clock/reset protection requested on brief p.3: not implemented (VERIFIED absence: one CTS tree).
- Cost of separation relative to the baseline (period 7.0 -> 7.2 ns, wirelength, power): numbers in 07_verification ([summary.md](#) lines 129-170).

5. Routing (summary)

Detailed evidence (STA, DRC/LVS scope, LEC, GLS) is in 07_verification; the routing-stage figures that describe the layout are:

Item	Value	Source	Conditions	Status
Global-route congestion	total usage 39.36 %, overflow 0 on every layer; met1 62.90 %, met2 40.23 %, met3 22.09 %, met4 12.33 %, met5 0.00 %; GRT wirelength 326452 um, 6833 routed nets	5_1_grt.log lines 138-152	FastRoute in OpenROAD (unknown)	VERIFIED

Item	Value	Source	Conditions	Status
Detailed route	TritonRoute "Number of violations = 0"; wirelength 244368 um (met1 118302, met2 73048, met3 43697, met4 9180, met5 138 um); 45501 vias; 5_route_drc.rpt is empty	5_2_route.log lines 1007, 1031-1038	DRT-0349: LEF58_ENCLOSURE rules skipped for mcon, via, via2, via3, via4 (lines 12-21); see discrepancy register (09_review_notes)	VERIFIED
Antenna	0 net / 0 pin violations after repair; 20 diodes	5_2_route.log lines 1056-1059; 6_report.log line 54	OpenROAD ANT checker only	VERIFIED
Clock tree	H-tree on clk, root/sink buffer clkbuf_16, 56 clock buffers created, 51 leaf buffers, 40 dummy loads, path depth 2-3, average sink wire length 596.74 um	4_1_cts.log lines 10-54	clock_tree_synthesis - sink_clustering_enable - repair_clock_nets	VERIFIED

6. Figures

6.1 KLayout views (klayout_views/)

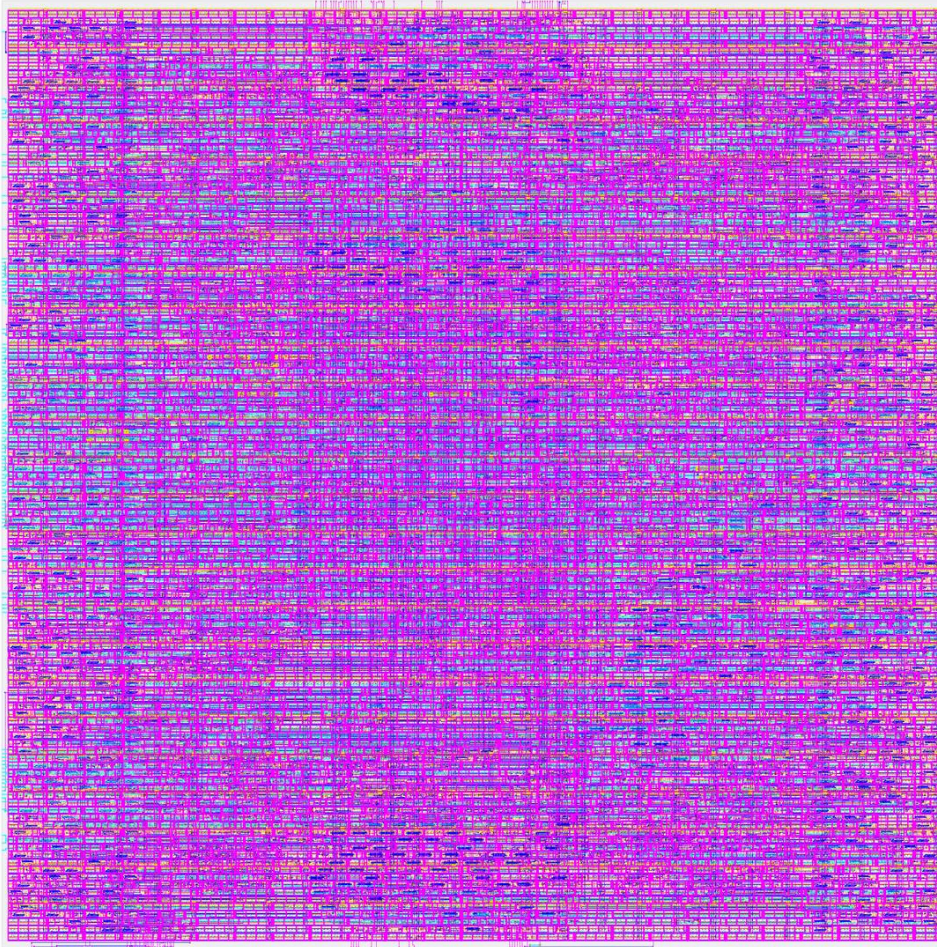
Rendered from the final GDS by [render_gds.py](#) (sha256 d3603e13...), KLayout 0.30.12 Python module (klayout.db, klayout.lay), layer colours from the ORFS sky130hd.lyp (build/pd_sep/platform/sky130hd.lyp, copied from the ORFS platform; not in the package), white background, text off. Command, from the repository root:

```
<python with klayout==0.30.12> -I review/orbit_demo_design_review/06_physical_design/klayout_views/
render_gds.py \
  build/pd_sep/results/sky130hd/orbit_demo/sep/6_final.gds \
  build/pd_sep/platform/sky130hd.lyp \
  reports/pdsep/regions.json \
  viz/layout.json \
  review/orbit_demo_design_review/06_physical_design/klayout_views
```

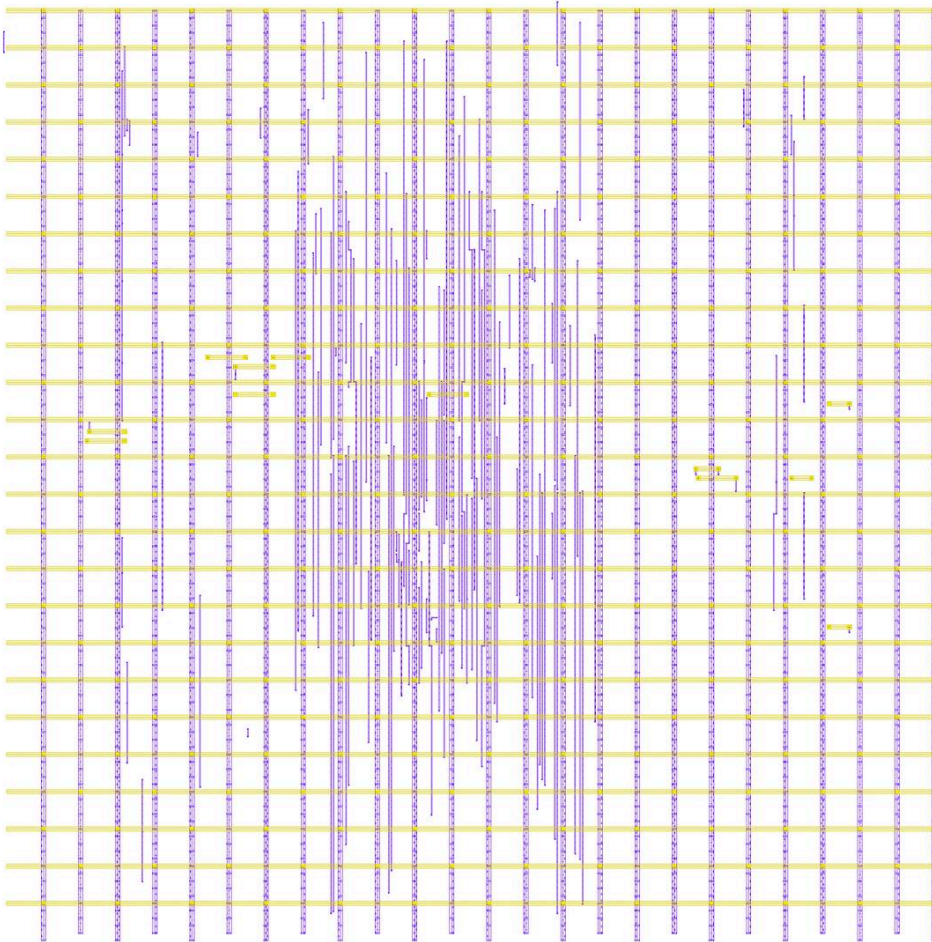
The command that wrote the package images was not logged (MISSING); the command above, run on 2026-10-06 into a scratch directory, reproduced all 10 PNGs byte-for-byte, and so did a second run with the package inputs (gunzipped [6_final.gds.gz](#) and [regions.json](#), which is byte-identical to reports/pdsep/regions.json) (**REPRODUCED 2026-10-06**). The black outlines in v03, v04 and v05 are a review overlay (layer 1000/0 added in memory, not written to any GDS) drawn from the fence boxes in regions.json; they are not layout geometry.

Layer sets used by the script (GDS layer/datatype, names from sky130hd.lyp):

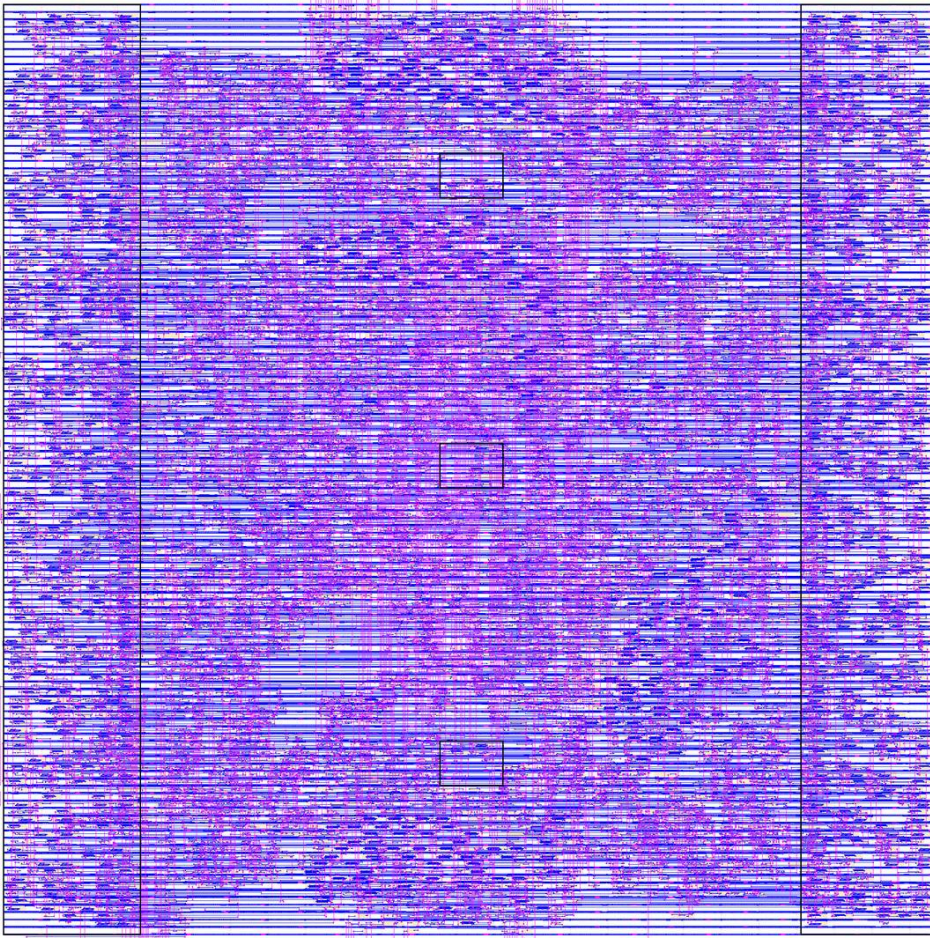
Set	Layers
all	every layer listed in sky130hd.lyp, default visibility
PDN	met4 71/20, via4 71/44, met5 72/20
LOW	li1 67/20, mcon 67/44, met1 68/20, via 68/44, met2 69/20
FEOL	nwell 64/20, diff 65/20, tap 65/44, poly 66/20, licon1 66/44, li1 67/20, mcon 67/44, met1 68/20, nsdm 93/44, psdm 94/20, npc 95/20



*v01_full_die_all_layers.png (2400 x 2400 px; window (-4, -4)-(350.295, 350.295) um): full die, all layers;
source 6_final.gds via render_gds.py.*



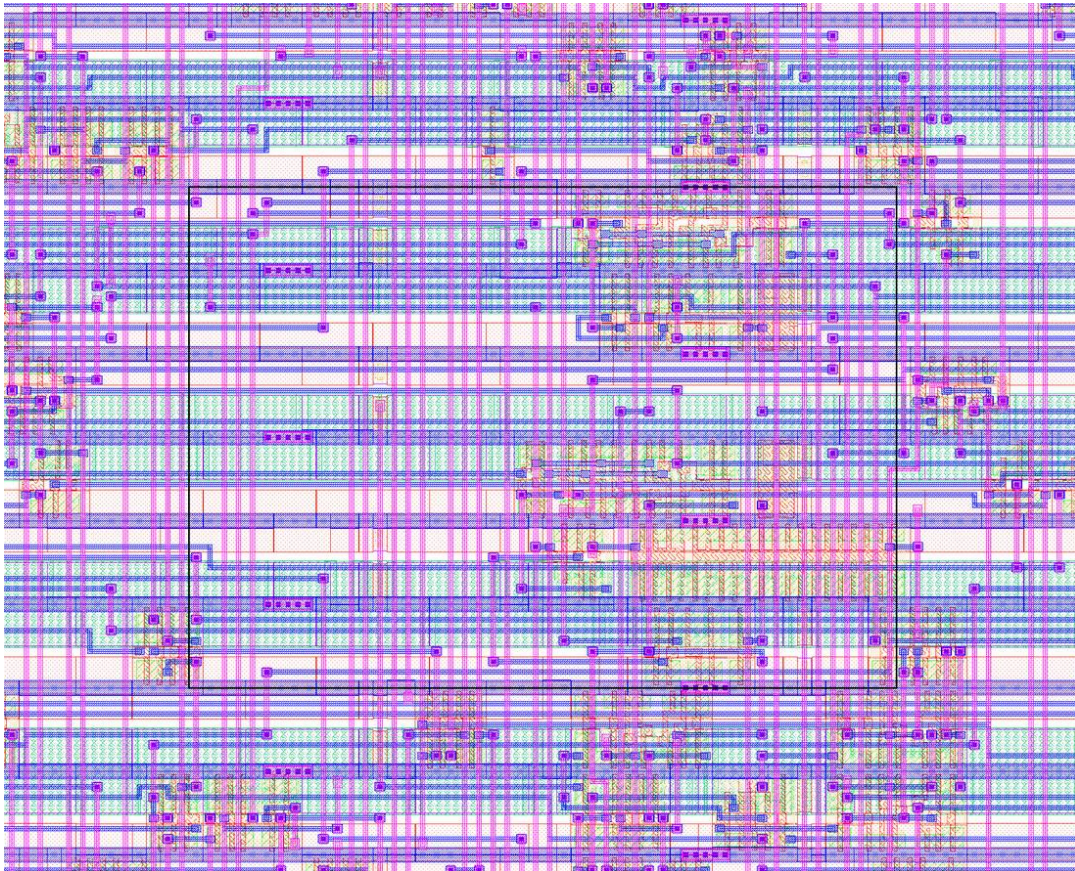
v02_full_die_pdn_met4_met5.png (2400 x 2400 px; full die): PDN set (met4, via4, met5): the 25 met4 and 25 met5 straps of section 3.1 plus the signal wires routed on met4 (9180 um) and met5 (138 um); source 6_final.gds.



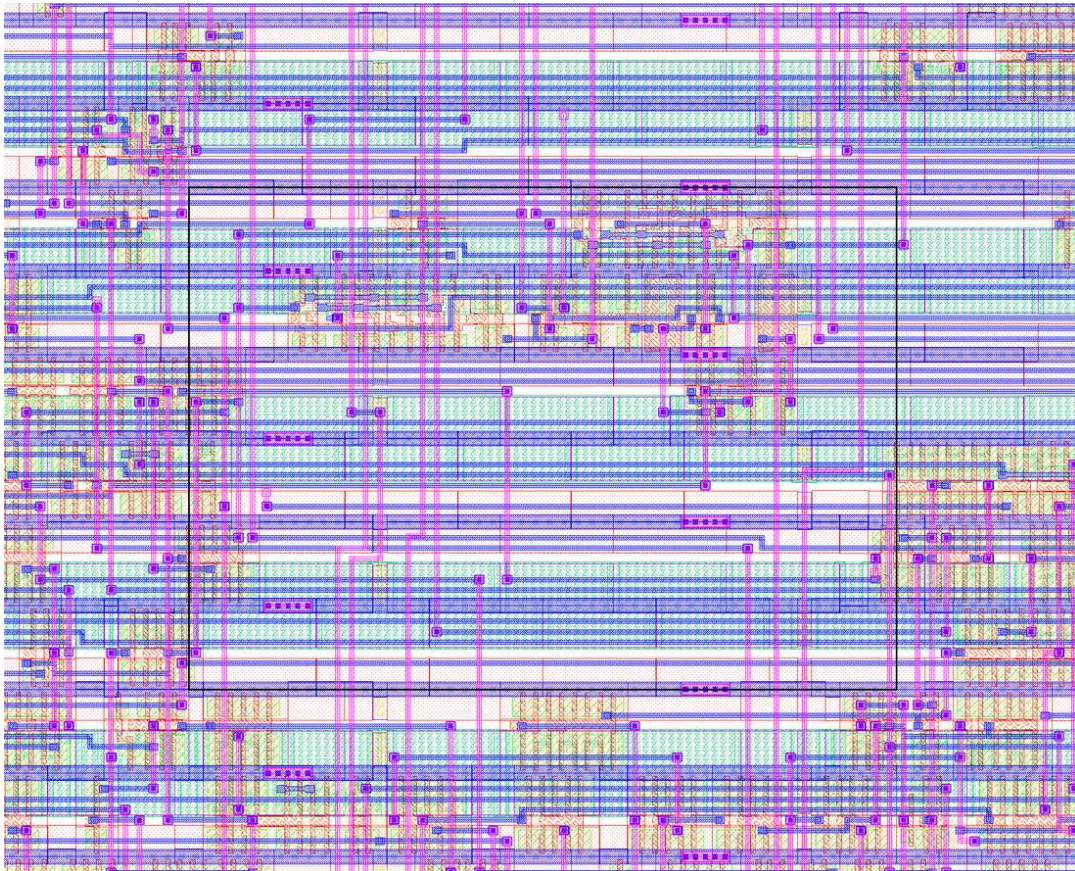
v03_full_die_li1_met1_met2_with_fence_overlay.png (2400 x 2400 px; full die): LOW set (li1, mcon, met1, via, met2) with the five fence boxes as a black overlay from regions.json; source 6_final.gds + regions.json.



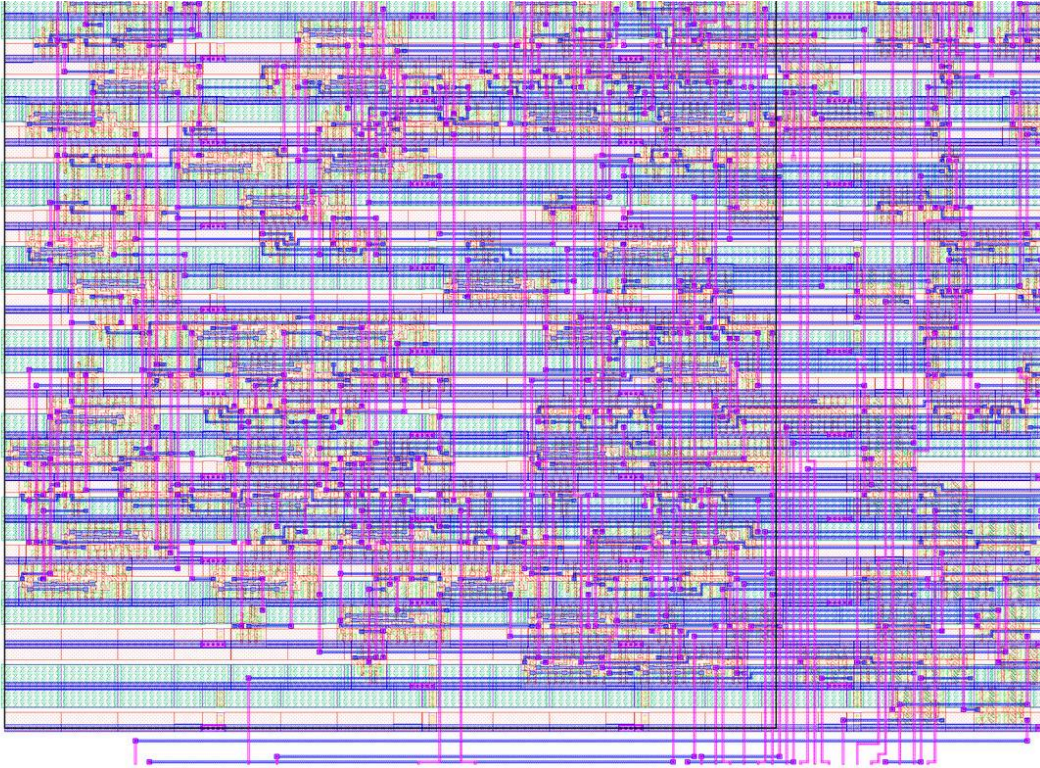
v04_detail_thermal_fence_th0.png (2000 x 1618 px; window (155.92, 266.00)-(190.92, 294.32) um = pdsep_th0 box + 6 um): FEOL + LOW sets, black overlay = pdsep_th0 box from regions.json; shows the 8 member cells of u_thermal.u_copy0 and the non-member cells around and overlapping the fence.



v04_detail_thermal_fence_th1.png (2000 x 1618 px; window (155.92, 159.92)-(190.92, 188.24) um): FEOL + LOW sets, overlay = pdsep_th1 box (u_thermal.u_copy1); clkbuf_0_clk overlaps this fence (summary.md line 123).



v04_detail_thermal_fence_th2.png (2000 x 1618 px; window (155.92, 51.12)-(190.92, 79.44) um): FEOL + LOW sets, overlay = pdsep_th2 box (u_thermal.u_copy2).



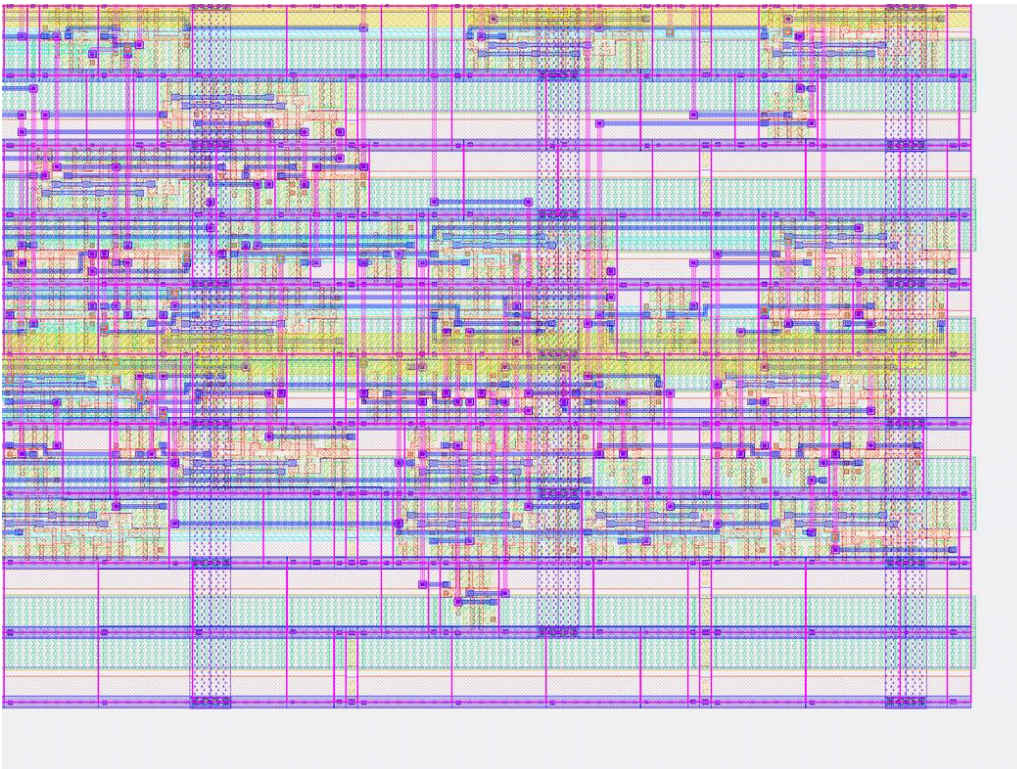
v05_detail_copyA_fence_lower_left.png (2400 x 1714 px; window (0, 0)-(70, 50) um): FEOL + LOW sets, overlay = pdsep_copyA box; lower-left corner of the copy-A fence with its right edge at x = 52.44 um and unconstrained logic to the right; source 6_final.gds + regions.json.



v06_detail_lane0_multiplier_region.png (2400 x 1481 px; window (123.33, 8.24)-(206.81, 59.76) um): LOW set, no overlay. The window is the core_um box of part L0.mult in viz/layout.json plus 4 um; see 6.3 for what that box is.



v07_detail_std_cell_rows_feol.png (2400 x 1440 px; window (150, 150)-(170, 162) um, about 4.4 rows): FEOL set; standard-cell rows with nwell, diffusion, poly, licon, li1, mcon and the met1 followpin rails.



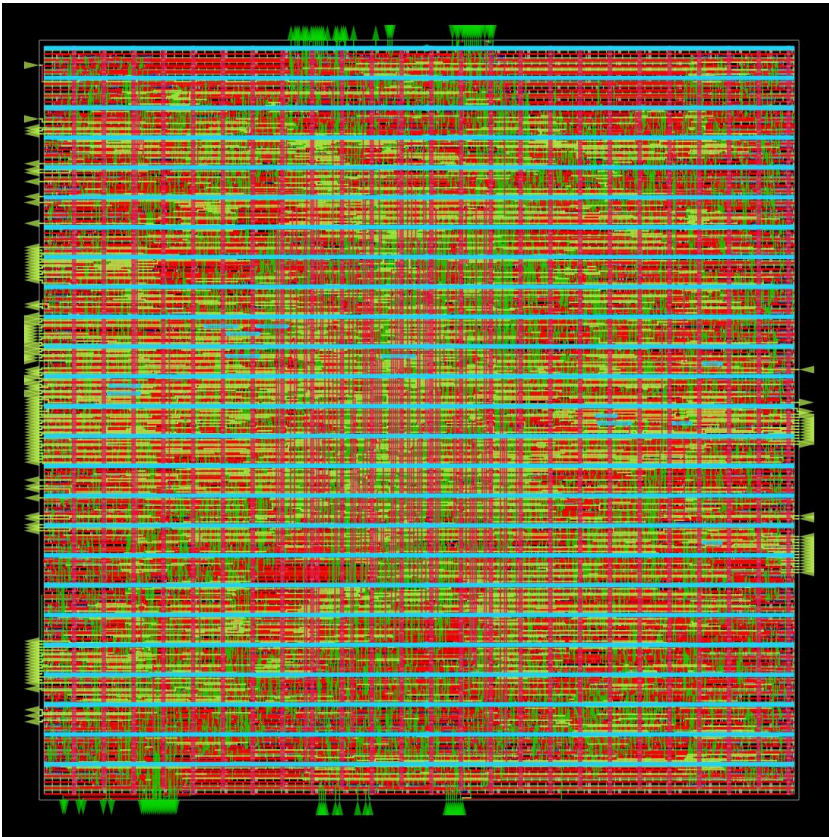
v08_detail_die_corner_lower_right_all_layers.png (2400 x 1828 px; window (306.295, -2)-(348.295, 30) um): all layers; lower-right die corner inside the copy-B fence area, with the core edge, the die boundary, met4 straps and the first met5 straps; no IO pins lie in this window (pinout.csv).

Also in the package: [orbit_demo_sky130hd_sep.png](#), a KLayout render with coloured fence outlines from regions.json written by [scripts/pdsep_render.py](#) (published report, 07_verification). Its label "fence gap 242 um" is the 241.50 um fence gap formatted to whole micrometres ({gap:.0f}, script line 151).

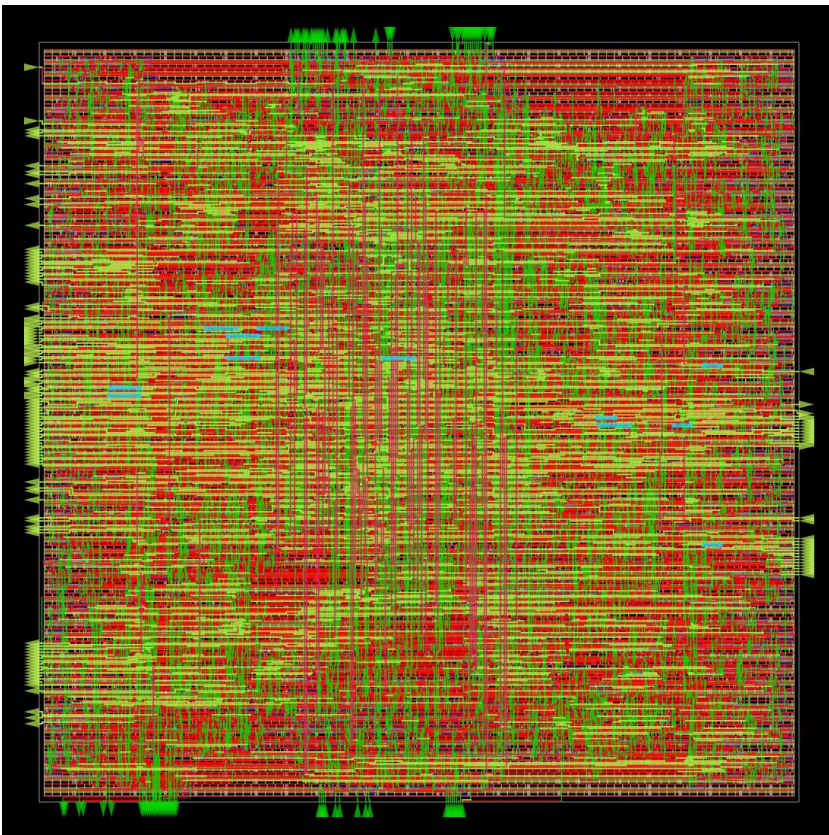
6.2 OpenROAD GUI images (orfs_images/)

Written by the ORFS final report step (save_images.tcl) at 22:09:44-45 on 2026-09-29 ([6_report.log](#) lines 68-77; the .webp.png names come from warning GUI-0010) and byte-identical to build/pd_sep/reports/sky130hd/orbit_demo/sep/* .webp.png (cmp, **REPRODUCED 2026-10-06**). OpenROAD version unknown, final database 6_final.odb, TT liberty. 1099 x 1099 px except cts_default_clk

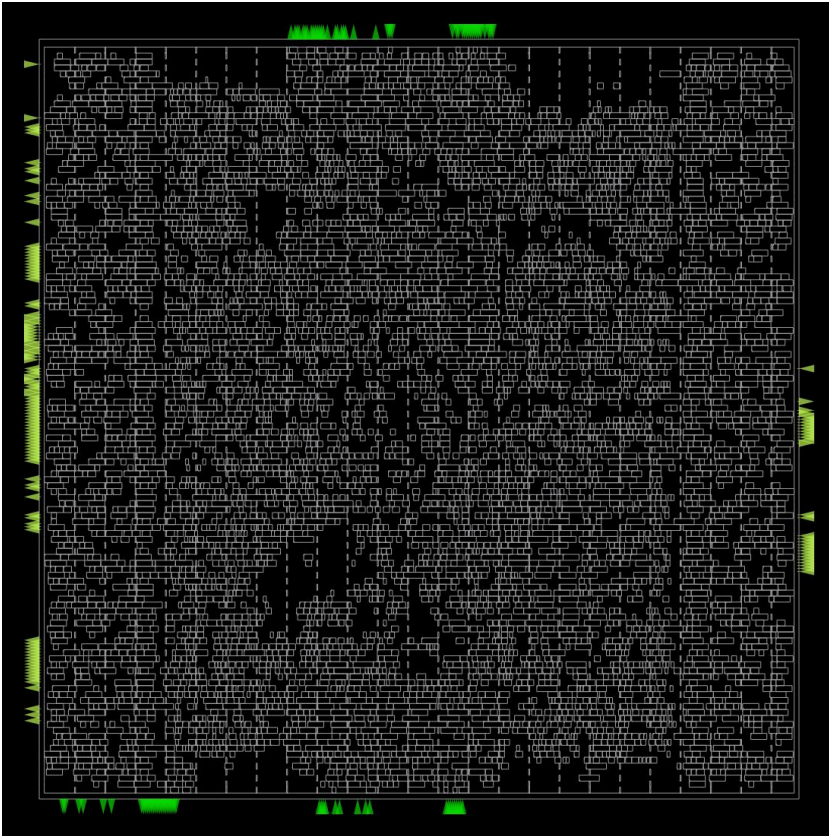
(1022 x 1022). The view descriptions follow flow/scripts/save_images.tcl in the image on this host (ASSUMPTION: same script as the run). Status of each image as tool output: **VERIFIED**.



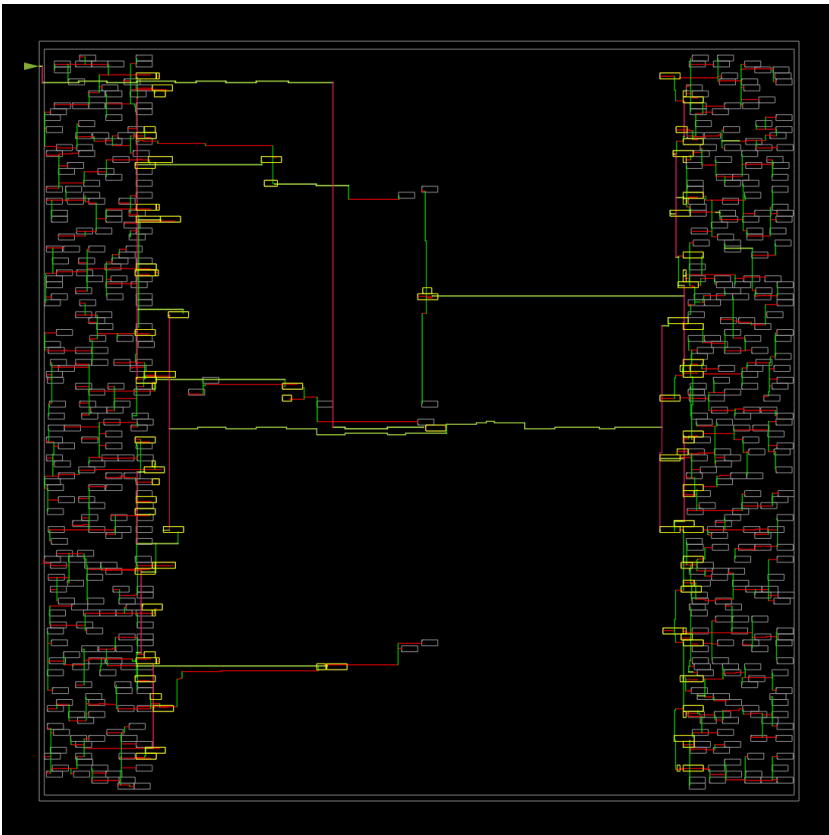
final_all.webp.png: routing view with all layers, nets including power/ground, instances and pins.



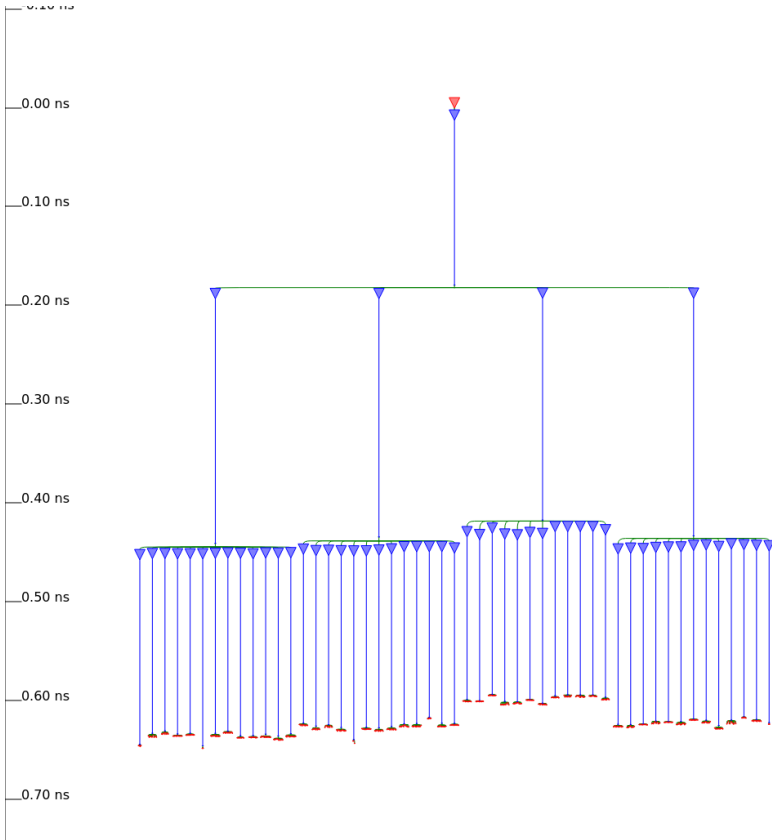
final_routing.webp.png: same view with the power and ground nets hidden (signal routing).



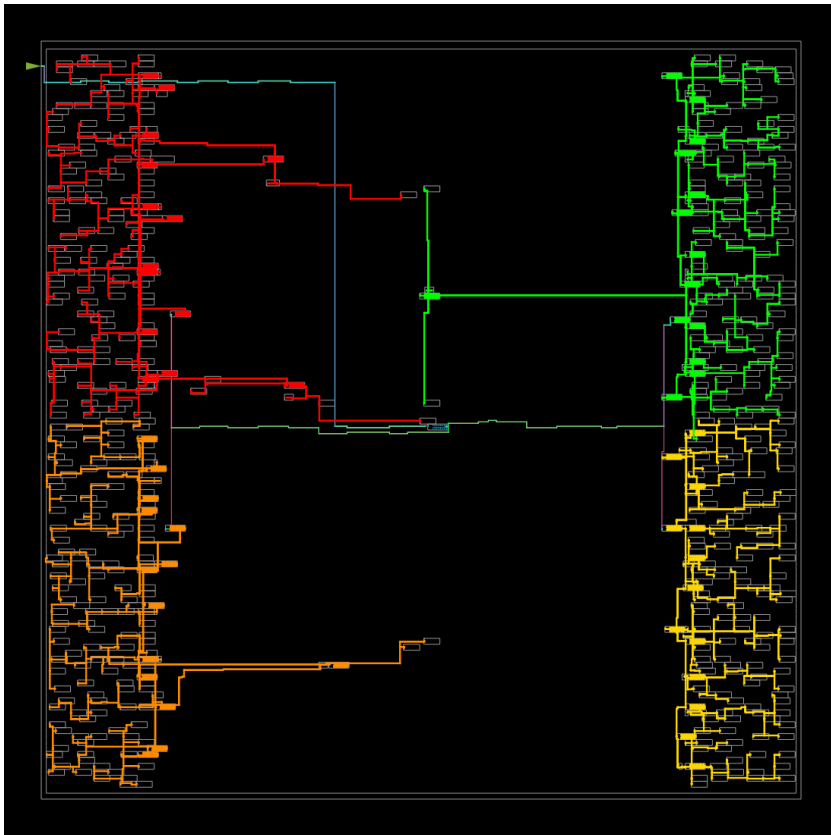
final_placement.webp.png: placement without routing shapes and without fill cells; the IO pins are drawn at the die edge.



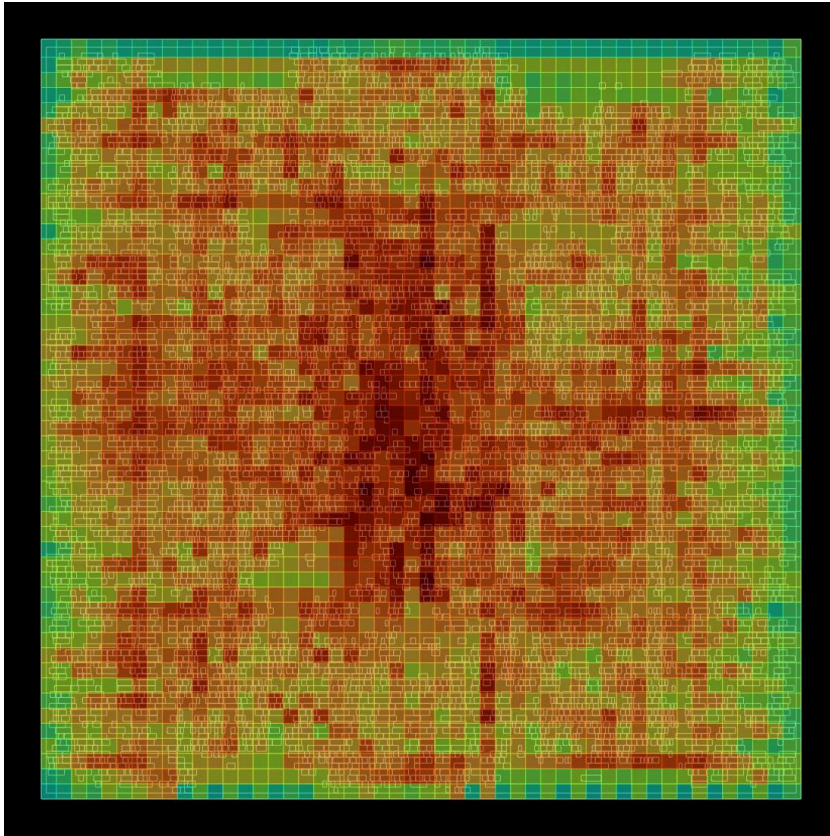
final_clocks.webp.png: clock nets, clock-tree cells and sequential cells only; most of the 521 sinks sit in the two copy fences at the left and right core edges, one tree reaches both (4_1_cts.log).



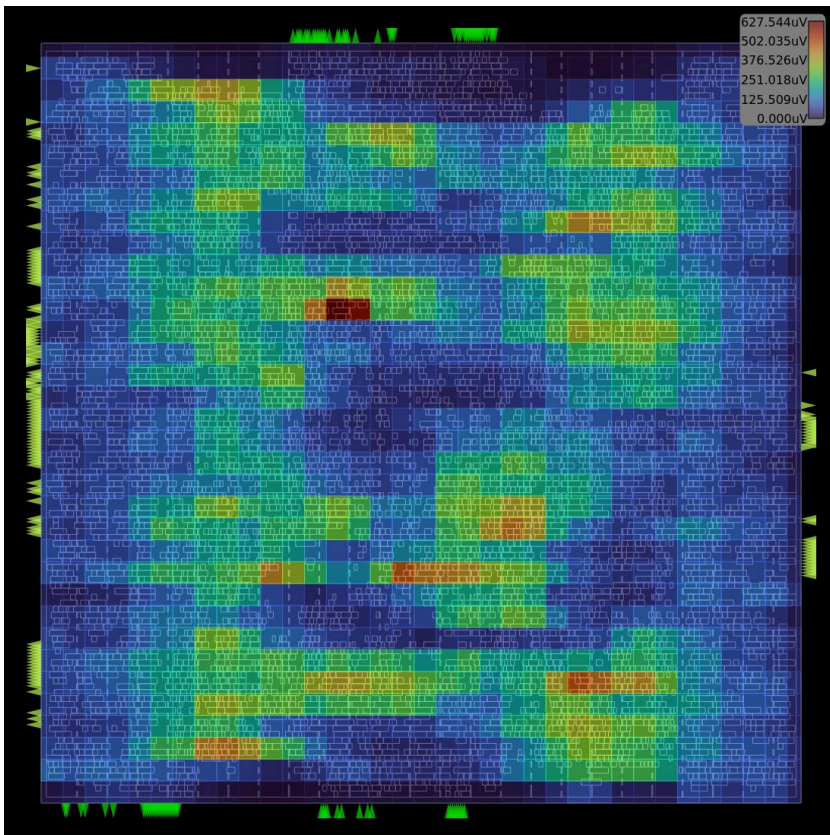
cts_default_clk.webp.png: Clock Tree Viewer for clock `clk`, scene default: insertion delay axis in ns, root buffer, four level-1 buffers, leaf buffers and sink arrivals (axis to 0.70 ns); final STA skew 0.0538 ns ([results.md](#) line 37).



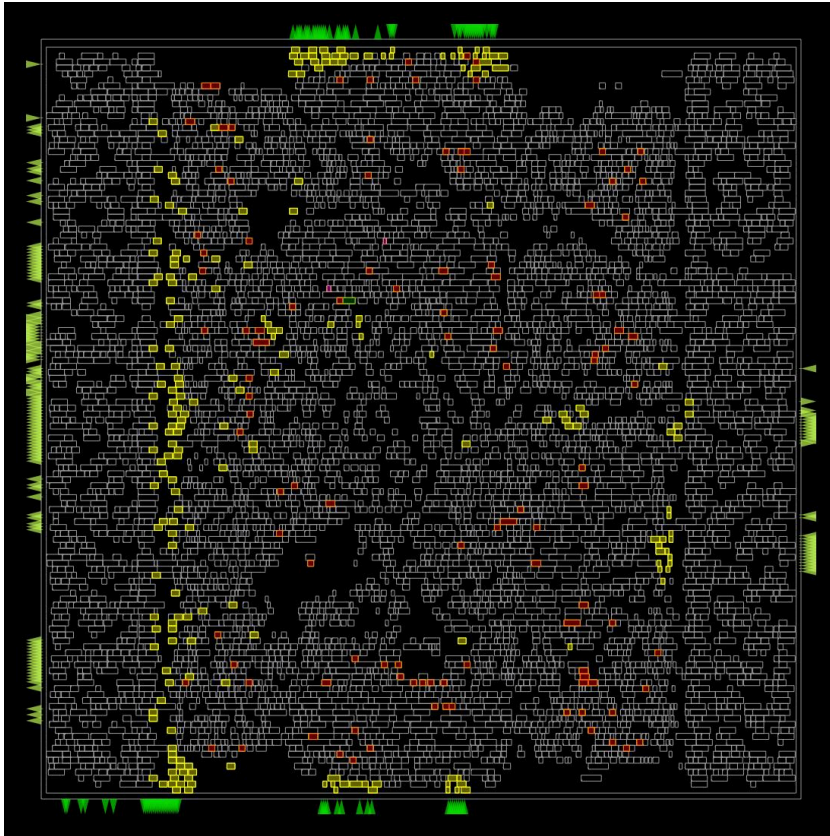
cts_default_clk_layout.webp.png: the same tree in the layout, coloured by branch.



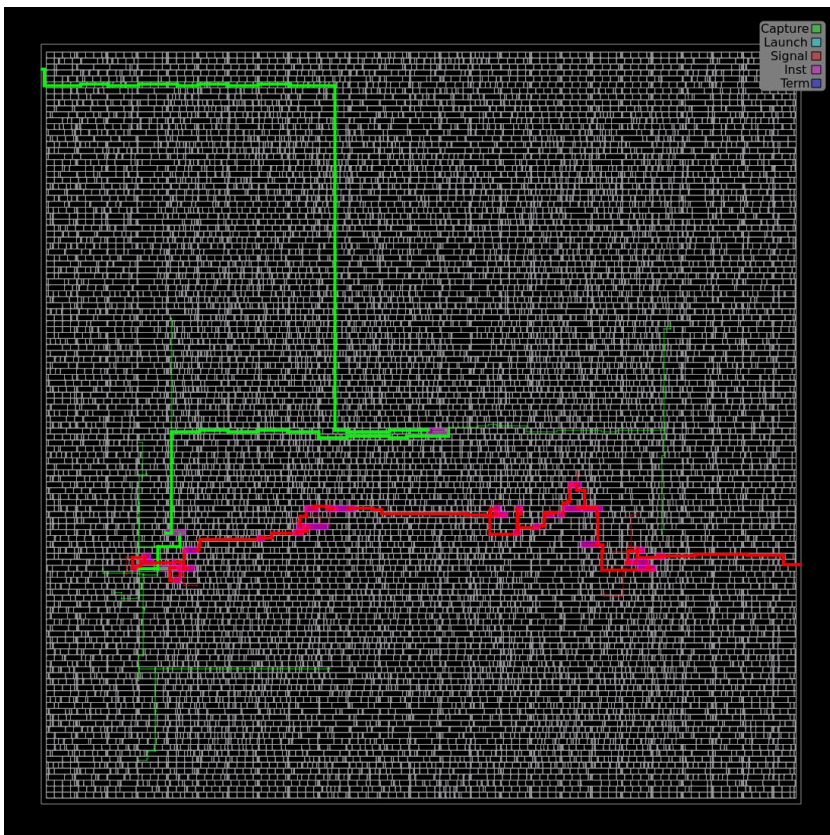
final_congestion.webp.png: routing-congestion heat map (no colour legend in the image); the GRT report gives 0 overflow on all layers (5_1_grt.log lines 138-148).



final_ir_drop.webp.png: static IR-drop heat map on layer met1 (platform IR_DROP_LAYER != met1); legend 0.000-627.544 uV. The net is not named in the image; the maximum equals the VSS worst-case 6.28e-04 V of 6_report.log line 48. Default-activity power, see 3.1.



final_resizer.webp.png: instances created by the resizer highlighted by name: input*/output* port buffers yellow, repeater*/fanout*/wire*/max_* magenta, rebuffer* red, split* dark green, hold* green (no hold cells exist in this run). The 867 place* repair buffers (count from the DEF) are not highlighted by that script.



final_worst_path.webp.png: worst setup path (gui::show_worst_path): data path (red) from in_a[27] on the E edge to g_lane[3].u_lane.u_res_a/q[29] in the copy-A fence; launch/capture clock paths in green; slack 0.03 ns at 7.2 ns, TT (*worst_setup_path.txt*).

6.3 Detailed views of blocks: what exists

View	Exists	Block box source	Status
Lane-0 multiplier region	v06	viz/layout.json part L0.mult ("Lane 0 / Multiplier", 197 cells, attributed by netlist cone: cells whose combinational fan-out reaches both copies of lane 0 and whose fan-in is only lane-0 operand inputs). core_um = 10th-90th percentile box of the cell centres, (127.33, 12.24)-(202.81, 55.76) um; the full extent bbox_um is (109.48, 5.44)-(214.36, 68.00) um, so the view shows the dense core of the multiplier, not all of its cells, and also contains cells of other parts (e.g. the lane-0 comparator L0.cmp, core box (139.56, 17.68)-(206.54, 93.02) um). viz/layout.json was derived from 6_final.def (inputs def_sha256 f5c544f3..., gds_sha256 cd19afa9..., both equal to the package files) by scripts/viz_parts.py (lines 711-751 for the boxes), generated 2026-09-29 23:02:30 UTC	REPRODUCED 2026-10-06 (hashes; the v06 window equals core_um $\pm$ 4 um); part attribution itself is CLAIM (unverified)
Thermal fences th0, th1, th2	v04 (three files)	regions.json fence boxes	REPRODUCED 2026-10-06
Copy-A fence corner	v05	regions.json pdsep_copyA box	REPRODUCED 2026-10-06
Standard-cell rows (FEOL)	v07	fixed window, no block	REPRODUCED 2026-10-06
Die corner	v08	fixed window at the die bbox	REPRODUCED 2026-10-06
Other blocks (lanes 1-3, adders L*.addA/addB, comparators L*.cmp, mismatch OR tree fault, thermal voter th.vote, clock root)	no view	boxes exist in viz/layout.json parts list (59 parts)	MISSING (render with render_gds.py using those core_um boxes if needed)

7. Open items for this section

Item	Status	Needed
ORFS commit, OpenROAD build version, open_pdks / skywater-pdk version, image digest in the run log	MISSING	record per run; pin the image by digest in mk/pdsep.mk
Platform scripts used by the run (pdn.tcl, tapcell.tcl, save_images.tcl)	ASSUMPTION (read from the current image)	copy them into the build directory at run time
Pinout specification	MISSING (pins are placer output)	specified pinout, then set_io_pin_constraint / pin file
Pad ring, IO cells, ESD, package	N/A for this core block (none designed)	padframe and package if the block is to be fabricated
Decap insertion and decoupling budget	MISSING (0 decap cells)	decision on decap fill, dynamic IR analysis
Metal fill and density check	MISSING (6_1_fill copies the ODB; KLayout deck has no density rules)	fill generation and density check on the filled GDS
Dynamic IR drop; EM check	MISSING (PSM EM currents at default activity computed 2026-10-06, not compared with the tlef current-density limits)	switching activity from simulation; comparison of the PSM currents with the DCCURRENTDENSITY limits
Physical basis of the 20 um / 10 um separation targets	MISSING	sky130 multi-node-upset / charge-sharing data
Detailed views of blocks other than lane-0 multiplier and the fences	MISSING	additional render windows

Item	Status	Needed
Command log of the original package renders	MISSING (reproduced byte-for-byte on 2026-10-06)	-